A horseshoe prior for sparse Optimal Multiple Co-Inertia Analysis: model, Gibbs sampler and discussion of the selection rule

1 Introduction

In the sparse version of OMCOA, the block weights $\sigma_k^{(r)}$ of $\mathbf{X}_k \approx \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \dots, \sigma_k^{(R)})$, are shrunk towards zero so that the pattern of non-zero weights reveals common, partially common, specific and inactive dimensions. A first solution uses a spike-and-slab prior, which puts a point mass at zero. An alternative is the horseshoe prior [1], a continuous global–local shrinkage prior, which was recently used on the singular values of a Bayesian SVD by Wang et al. [4]. This note adapts their approach to OMCOA, gives the corresponding Gibbs sampler, and discusses the main practical issue raised by a continuous prior in this context: the rule used to decide which weights are non-zero.

2 Model

Let $\mathbf{X}_1, \dots, \mathbf{X}_K$ be column-centered blocks with $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$ and $R \leq \min_k p_k$ a maximal rank:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \text{ with i.i.d. entries } \mathcal{N}(0, \tau_k^{-1}), \tag{1}$$

with $\mathbf{U}^\intercal\mathbf{U} = \mathbf{I}_R$, $\mathbf{1}_N^\intercal\mathbf{U} = 0$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = \mathbf{I}_R$. Following [4], the weights receive a horseshoe prior truncated to $(0, \infty)$ and scaled by the noise level of their block:

$$\sigma_k^{(r)} \mid \eta_k^{(r)}, \zeta, \tau_k \;\sim\; \mathcal{N}^+\Big(0,\; \frac{(\eta_k^{(r)})^2\zeta^2}{\tau_k}\Big), \qquad \eta_k^{(r)} \sim \mathcal{C}^+(0,1), \qquad \zeta \sim \mathcal{C}^+(0,1), \tag{2}$$

where $\mathcal{C}^+(0,1)$ is the standard half-Cauchy distribution. The local scales $\eta_k^{(r)}$ let individual weights escape shrinkage, while the global scale ζ adapts the overall level of shrinkage to the number and size of the signals. As in the non-sparse sampler, $\mathbf{U}$ and $\mathbf{V}_k$ are uniform on their Stiefel manifolds and $\tau_k \sim \mathrm{Gamma}(a_0, b_0)$.

Auxiliary variables. The half-Cauchy distribution is a scale mixture of inverse-gamma distributions [3]: if $x^2 \mid a \sim \mathrm{IG}(1/2, 1/a)$ and $a \sim \mathrm{IG}(1/2, 1/A^2)$, then $x \sim \mathcal{C}^+(0, A)$. Hence

$$(\eta_k^{(r)})^2 \mid \nu_k^{(r)} \sim \mathrm{IG}\big(1/2, 1/\nu_k^{(r)}\big), \qquad \nu_k^{(r)} \sim \mathrm{IG}(1/2, 1),$$
$$\zeta^2 \mid \xi \sim \mathrm{IG}(1/2, 1/\xi), \qquad \xi \sim \mathrm{IG}(1/2, 1),$$

and all the full conditional distributions are standard.

3 Full conditional distributions

With $d_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}$ and the decomposition $\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\sum_r \sigma_k^{(r)} d_k^{(r)} + \sum_r (\sigma_k^{(r)})^2$:

Proposition 1. *1.* $\mathbf{U}$ *and* $\mathbf{V}_k$*: matrix von Mises–Fisher distributions, as in the non-sparse sampler, with parameters* $\mathbf{Q}^\intercal \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k$ *for* $\tilde{\mathbf{U}}$ *(where* $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$*) and* $\tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k$ *for* $\mathbf{V}_k$*.*

2. Weights: with $\omega_k^{(r)} = \tau_k\big(1 + 1/((\eta_k^{(r)})^2\zeta^2)\big)$*,*

$$\sigma_k^{(r)} \mid \cdot \;\sim\; \mathcal{N}^+\Big((1-\kappa_k^{(r)})\, d_k^{(r)},\; \frac{1}{\omega_k^{(r)}}\Big), \qquad \kappa_k^{(r)} = \frac{1}{1 + (\eta_k^{(r)})^2\zeta^2} \in (0,1).$$

3. Local and global scales:

$$(\eta_k^{(r)})^2 \mid \cdot \sim \mathrm{IG}\Big(1,\; \frac{1}{\nu_k^{(r)}} + \frac{\tau_k(\sigma_k^{(r)})^2}{2\zeta^2}\Big), \qquad \nu_k^{(r)} \mid \cdot \sim \mathrm{IG}\Big(1,\; 1 + \frac{1}{(\eta_k^{(r)})^2}\Big),$$

$$\zeta^2 \mid \cdot \sim \mathrm{IG}\Big(\frac{KR+1}{2},\; \frac{1}{\xi} + \sum_{k,r} \frac{\tau_k(\sigma_k^{(r)})^2}{2(\eta_k^{(r)})^2}\Big), \qquad \xi \mid \cdot \sim \mathrm{IG}\Big(1,\; 1 + \frac{1}{\zeta^2}\Big).$$

4. Noise precisions (the prior (2) *depends on* τ_k*):*

$$\tau_k \mid \cdot \sim \mathrm{Gamma}\Big(a_0 + \frac{(N-1)p_k + R}{2},\; b_0 + \frac{\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2}{2} + \sum_r \frac{(\sigma_k^{(r)})^2}{2(\eta_k^{(r)})^2\zeta^2}\Big).$$

The quantity $\kappa_k^{(r)}$ is the *shrinkage factor*: the conditional mean of $\sigma_k^{(r)}$ is $(1-\kappa_k^{(r)})d_k^{(r)}$, so that $\kappa_k^{(r)} \approx 1$ means that the weight is shrunk to zero and $\kappa_k^{(r)} \approx 0$ that it is left almost untouched.

Algorithm 1 Horseshoe Gibbs sampler for sparse OMCOA

Require: centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$; maximal rank R; iterations T; burn-in T_0; a_0, b_0

Initialization: $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ from OMCOA with R dimensions; $\eta_k^{(r)} = \nu_k^{(r)} = \zeta = \xi = 1$; $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \dots, T$ **do**
  update $\mathbf{U}$, then each $\mathbf{V}_k$, by one column-wise sweep of their von Mises–Fisher conditionals
  for $k = 1, \dots, K$ **do**
    draw $\sigma_k^{(1)}, \dots, \sigma_k^{(R)}$ from their truncated normal conditionals; draw τ_k
  end for
  draw $(\eta_k^{(r)})^2$, $\nu_k^{(r)}$ for all (k, r), then ζ^2 and ξ
  if $t > T_0$ **then** store $\mathbf{U}$, Σ_k, $\kappa_k^{(r)}$, τ_k
  end if
end for
align the stored draws (permutation and signs of the dimensions) on a reference $\mathbf{U}_{\mathrm{ref}}$

4 Simulation

We used the same design as for the spike-and-slab sampler: $K = 3$, $N = 100$, $(p_1, p_2, p_3) = (10, 5, 15)$, 10% of noise, true rank 3 with

$$\Sigma_1 = \mathrm{diag}(0.7,\, 0.5,\, 0), \qquad \Sigma_2 = \mathrm{diag}(0.6,\, 0.4,\, 0), \qquad \Sigma_3 = \mathrm{diag}(0.5,\, 0,\, 0.6),$$

(dimension 1 common, dimension 2 shared by blocks 1 and 2, dimension 3 specific to block 3), and a maximal rank $R = 4$ ($T = 6000$, $T_0 = 2000$). Table 1 reports, for each weight, the posterior mean, the quantity $1 - \mathrm{E}[\kappa_k^{(r)} \mid \mathbf{X}]$, and the posterior probability that the dimension explains more than 1% of the variance of the block (Section 5).

Table 1: Horseshoe sampler: posterior summaries of the weights (true non-zero entries in bold).

Block	Dim	True $\sigma_k^{(r)}$	$\mathrm{E}[\sigma_k^{(r)} \mid \mathbf{X}]$	$1 - \mathrm{E}[\kappa_k^{(r)} \mid \mathbf{X}]$	$P(s_k^{(r)} > 1\% \mid \mathbf{X})$
1	1	**0.7**	0.710	1.000	1.000
1	2	**0.5**	0.511	0.999	1.000
1	3	0	0.011	0.804	0.000
1	4	0	0.021	0.853	0.000
2	1	**0.6**	0.600	0.999	1.000
2	2	**0.4**	0.392	0.999	1.000
2	3	0	0.015	0.793	0.000
2	4	0	0.020	0.807	0.004
3	1	**0.5**	0.499	1.000	1.000
3	2	0	0.008	0.756	0.000
3	3	**0.6**	0.602	1.000	1.000
3	4	0	0.019	0.869	0.000

As an estimator, the horseshoe behaves as expected: the non-zero weights are estimated without visible bias, and the null weights are shrunk to values between 0.008 and 0.021, i.e. one to two orders of magnitude below the signals. The question is how to turn these continuous estimates into a decision.

5 Discussion: which selection rule?

5.1 The standard rule fails

In the normal-means problem $y \sim \mathcal{N}(\beta, 1)$, Carvalho et al. [1] declare β non-zero when $1 - \mathrm{E}[\kappa \mid y] > 1/2$, i.e. when the posterior mean keeps more than half of the observation. Applied to OMCOA, this rule declares *every* weight active (Table 1): the null weights have $1 - \mathrm{E}[\kappa]$ between 0.76 and 0.87.

The reason is specific to SVD-type models. The rule is calibrated for an observation equal to the parameter plus a standard noise. In OMCOA, the "observation" of $\sigma_k^{(r)}$ is $d_k^{(r)} = \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$, and the directions $\mathbf{u}^{(r)}$ and $\mathbf{v}_k^{(r)}$ are themselves fitted to the data. For a null weight, $\mathbf{v}_k^{(r)}$ is chosen on a sphere of dimension $p_k - R$ so as to maximize $d_k^{(r)}$: it aligns with the largest noise direction available, and $d_k^{(r)}$ is of the order of $\sqrt{p_k - R + 1}$ noise standard deviations, or more when $\mathbf{u}^{(r)}$ is free as well. Measured in units of the noise standard deviation, a null weight therefore looks like a small

but genuine signal, and the horseshoe, correctly, does not shrink it completely. In other words, the shrinkage factor $\kappa_k^{(r)}$ is a relevant measure of shrinkage, but 1/2 is not a relevant threshold here.

This is consistent with Wang et al. [4], who do not use κ to select the rank of their Bayesian SVD, but the ratio of consecutive posterior singular values (see below).

5.2 Alternative rules

Explained variance. A natural criterion in multiblock analysis is the share of the variance of block k explained by dimension r, $s_k^{(r)} = (\sigma_k^{(r)})^2/\|\mathbf{X}_k\|^2$. A weight is declared active when $P(s_k^{(r)} > c \mid \mathbf{X}) > 1/2$ for a threshold c fixed by the user. With $c = 1\%$, the structure is recovered exactly (last column of Table 1). The separation is wide: the posterior mean of $s_k^{(r)}$ is at most 0.12% for the null weights and at least 27% for the non-zero ones, so the result is not sensitive to the choice of c in a broad range. The drawback is that c is a substantive choice rather than a statistical one.

Ratio of consecutive weights. Following [4], the posterior means of the weights of each block can be sorted in decreasing order, $\hat{\sigma}_{k,(1)} \geq \cdots \geq \hat{\sigma}_{k,(R)}$, and the number of active dimensions of block k taken as $\hat{R}_k = \arg\max_j \hat{\sigma}_{k,(j)}/\hat{\sigma}_{k,(j+1)}$. In the simulation the largest ratios are 24.6, 20.0 and 26.4 for blocks 1, 2 and 3, all located between the second and third largest weights, so that $\hat{R}_k = 2$ for each block and the structure is recovered. This scree-type rule requires no threshold, but it always declares at least one active dimension per block and cannot conclude that all R dimensions are active; it is also less natural when R is small.

Noise floor. The argument above suggests a calibrated threshold: compare $\sigma_k^{(r)}$ with the value expected for a null weight, of the order of $c_k = \sqrt{(p_k - R + 1)/\tau_k}$, and declare the weight active when $P(\sigma_k^{(r)} > \rho\, c_k \mid \mathbf{X}) > 1/2$ for a constant $\rho > 1$. This rule adapts to the noise level and to the dimension of each block; its calibration (choice of ρ, e.g. by permutation of the rows of each block) has not been studied yet.

5.3 Horseshoe versus spike-and-slab

Both priors estimate the non-zero weights equally well in our simulation. They differ in how the structure is obtained (Table 2). The spike-and-slab prior answers the question of interest directly, through the probability that dimension r is present in block k, at the price of choosing the slab precision. The horseshoe avoids this choice and mixes more easily, but its continuous nature moves the selection problem to a post-processing rule, and the standard rule based on κ is not calibrated for SVD-type models. For the identification of common, partially common and specific structures, the spike-and-slab prior is therefore the more natural choice; the horseshoe is attractive when the goal is primarily the estimation of the weights, or as a robustness check of the structure selected by the spike-and-slab sampler.

References

[1] C. M. Carvalho, N. G. Polson and J. G. Scott (2010). The horseshoe estimator for sparse signals. *Biometrika*, 97(2), 465–480.

Table 2: Comparison of the two sparse priors for OMCOA.

	Spike-and-slab	Horseshoe
Exact zeros	yes	no (continuous shrinkage)
Selection	posterior inclusion probabilities, threshold 1/2	requires an additional rule (explained variance, ratio, noise floor)
Structure common / specific	read directly from $\gamma_k^{(r)}$	deduced from the selection rule
Tuning	slab precision λ (Lindley's paradox)	no slab scale; threshold of the selection rule
Mixing	switched-off dimensions are hard to switch on again	smoother (no discrete moves)
Uncertainty on the structure	yes ($P(\gamma_k^{(r)} = 1 \mid \mathbf{X})$)	indirect ($P(s_k^{(r)} > c \mid \mathbf{X})$)
Estimation of the weights	good	good, with adaptive shrinkage

[2] P. D. Hoff (2007). Model averaging and dimension selection for the singular value decomposition. *Journal of the American Statistical Association*, 102(478), 674–685.

[3] E. Makalic and D. F. Schmidt (2016). A simple sampler for the horseshoe estimator. *IEEE Signal Processing Letters*, 23(1), 179–182.

[4] J. Wang, Y. Gong, H. Lin, R. Zhang and W. Zhao (2025). Bayesian singular value decomposition procedure based on the horseshoe prior. *Journal of Applied Statistics*, doi:10.1080/02664763.2025.2589992.